

RicoSLAM: Long-Term Localization and Mapping in Dynamic Environments

Ricardo B. Sousa

Thesis Research Plan – Doctoral Program in Electrical and Computers Engineering

Zoom Meeting, January 12, 2023

Prof. António Paulo Moreira, PhD (Supervisor)

Héber Miguel Sobreira, PhD (Second Supervisor)

Outline

- Introduction
- Literature Review
- Thesis Statement
- Preliminary Developments
- Research Plan

Outline

- Introduction
 - Context
 - Motivations
 - Goals
- Literature Review
- Thesis Statement
- Preliminary Developments
- Research Plan

Introduction: Context

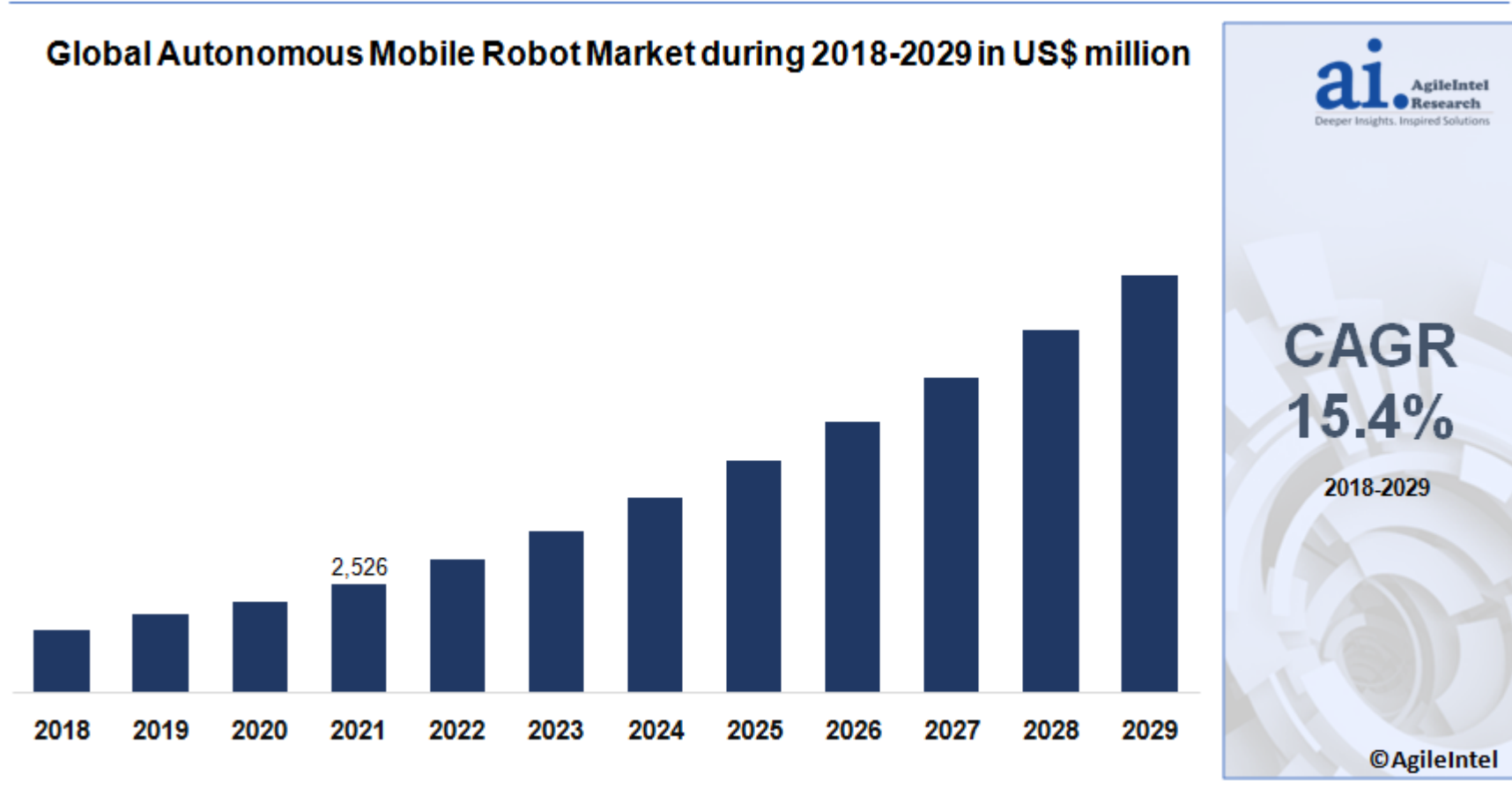


Fig. 1: Global Autonomous Mobile Robot (AMR) market estimated growth by AgileIntel Research during 2018–2029 in US\$ million [1]

Context

- Autonomous Mobile Robots (AMRs) requires an environment representation
- Simultaneous Localization and Mapping (SLAM)
 - Robot pose estimation
 - Mapping the environment

Context

- Autonomous Mobile Robots (AMRs) requires an environment representation
- Simultaneous Localization and Mapping (SLAM)
 - Robot pose estimation
 - Mapping the environment
- Static scene only requires mapping once the environment
- Real world is not static!
 - Moving elements
 - Environment reconfiguration
 - Appearance variations

Context

- Autonomous Mobile Robots (AMRs) requires an environment representation
 - Simultaneous Localization and Mapping (SLAM)
 - Robot pose estimation
 - Mapping the environment
 - Static scene only requires mapping once the environment
 - Real world is not static!
 - Moving elements
 - Environment reconfiguration
 - Appearance variations
-
- How to update the map?
 - Only most current state?
 - Only static elements?
 - When?
 - When changes occur?
 - After a certain time?

Motivation

- Strategic Research, Innovation and Deployment Agenda (SRIDA) (September 2020) [2]
 - Deal with unpredictable and unknown environments, and irreversible actions
 - Avoid unrecoverable states that require the reset of AMRs system

Motivation

- Strategic Research, Innovation and Deployment Agenda (SRIDA) (September 2020) [2]
 - Deal with unpredictable and unknown environments, and irreversible actions
 - Avoid unrecoverable states that require the reset of AMRs system
- Dynamic elements can make map representations outdated and unusable
 - Changes in the scene not updated in the map
 - Outdated map would require a specialized operator for remapping the scene



Motivation

- Strategic Research, Innovation and Deployment Agenda (SRIDA) (September 2020) [2]
 - Deal with unpredictable and unknown environments, and irreversible actions
 - Avoid unrecoverable states that require the reset of AMRs system
- Dynamic elements can make map representations outdated and unusable
 - Changes in the scene not updated in the map
 - Outdated map would require a specialized operator for remapping the scene
- Interpretation and dealing with dynamic changes online would improve the efficiency and self-awareness of those systems

Goals

- Modeling of environment dynamics perceived by different sensors
- Incorporate the modeled dynamics in both localization and mapping
- Lifelong SLAM in dynamic environments with an AMR

Outline

- Introduction
- Literature Review
 - Methodology
 - SLAM
 - Long-term SLAM in Dynamic Environments
 - Open Challenges
- Thesis Statement
- Preliminary Developments
- Research Plan

Literature Review: Methodology

- Search base string
`(robot* OR vehicle*) AND
 ((locali* AND map*) OR "slam") AND
 ("long term" OR "life long" OR
 lifelong)`
- Last full inquiry on May 17, 2022
- Selection Process
 1. Identification
 2. Screening
 3. Quality Assessment

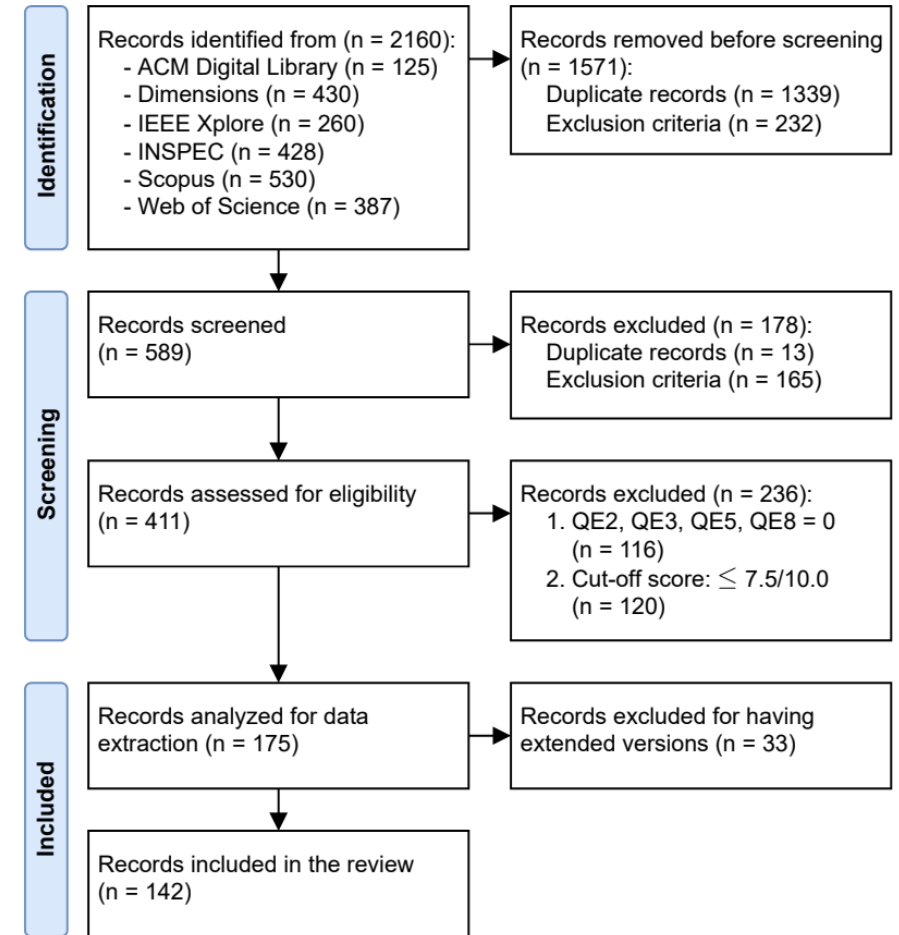


Fig. 2: PRISMA flow diagram

Literature Review: Methodology

- Search base string
 $(robot* \text{ OR } vehicle*) \text{ AND } ((locali* \text{ AND } map*) \text{ OR } "slam") \text{ AND } ("long \text{ term}" \text{ OR } "life \text{ long}" \text{ OR } lifelong)$
- Last full inquiry on May 17, 2022
 - Update on October 25, 2022, for dynamic environments
- Selection Process
 1. Identification
 2. Screening
 3. Quality Assessment

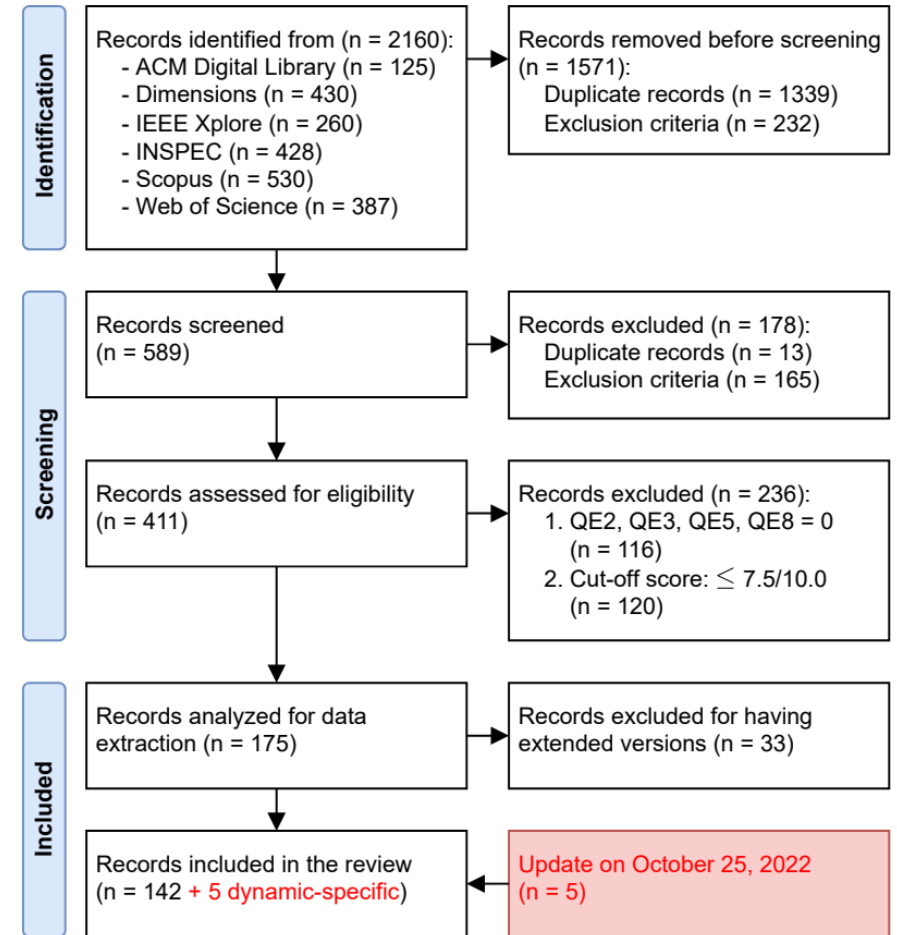


Fig. 2: PRISMA flow diagram

SLAM

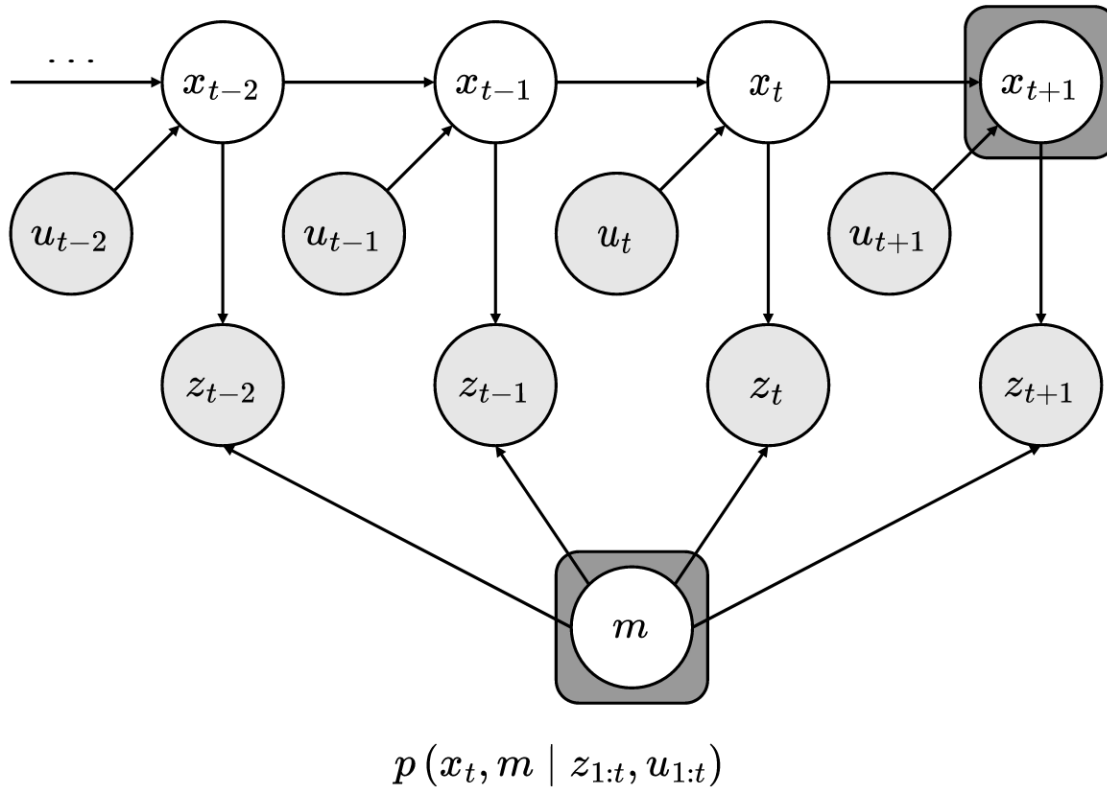


Fig. 3: Online SLAM problem [4]

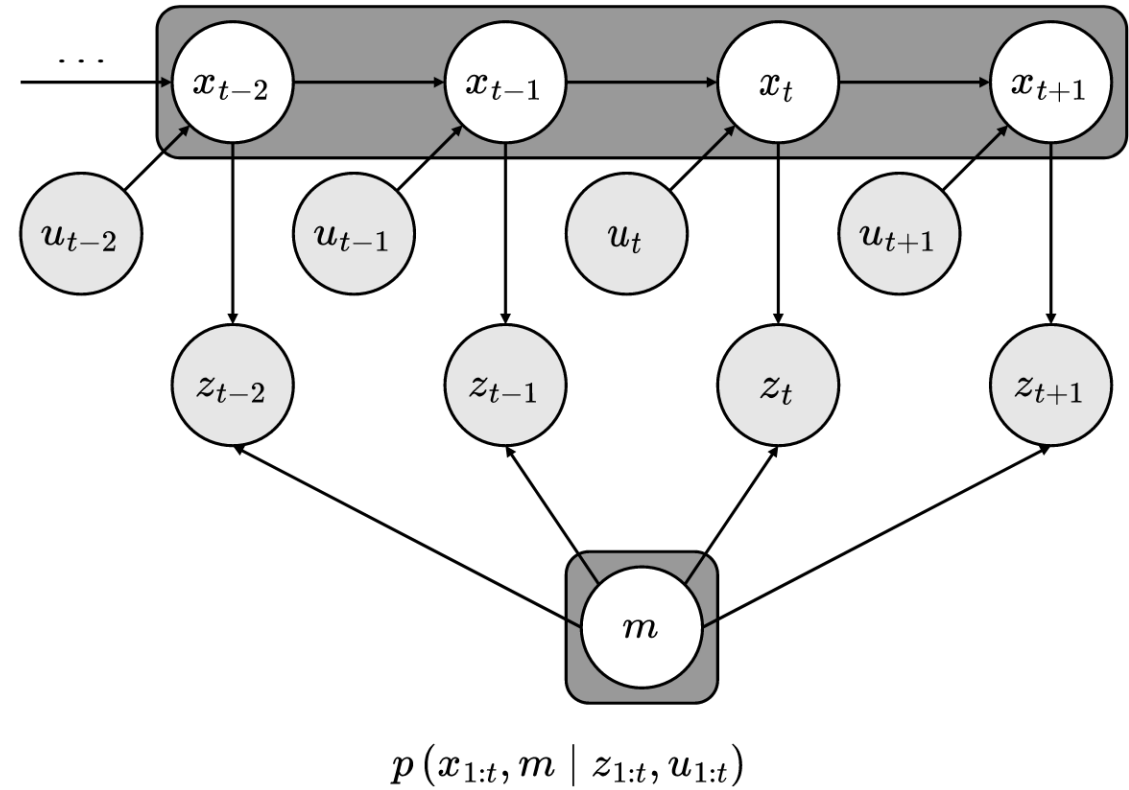


Fig. 4: Full SLAM problem [4]

SLAM

- Scan matching
- Particle filter
- Pose graph

SLAM

- Scan matching
 - Align sensor data with current map
 - Update the map with the latest sensor scan
- Particle filter
 - Rao-Blackwellized Particle Filters (RBPFs)
 - Probabilistic approach
 - SLAM = robot localization + map estimation conditioned on robot pose estimate
- Pose graph
 - Front (identify constraints) and back-end (refine graph) modules
 - Optimization approach
 - Maximum Likelihood Estimation (MLE) formulation for optimizing the pose graph

Long-term SLAM in Dynamic Environments

- Map representation
 - Short/Long-Term Memory (S/LTM)
 - Static vs dynamic maps
 - Multi-hypothesis framework

Long-term SLAM in Dynamic Environments

- Map representation
 - Short/Long-Term Memory (S/LTM)
 - Multi-store model of human memory
 - Multiple timescales
 - Use LTM maps for localization
 - Static vs dynamic maps
 - Multi-hypothesis framework

MULTI-STORE MODEL

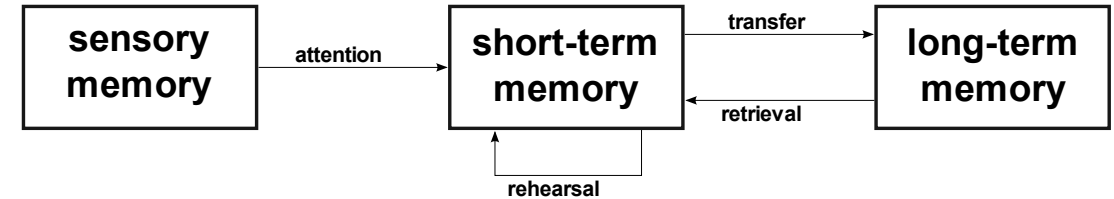


Fig. 5: Multi-store model of human memory proposed by Atkinson and Shiffrin (1968)

Source:

https://commons.wikimedia.org/wiki/File:Memory_multistore_model.svg

Long-term SLAM in Dynamic Environments

- Map representation
 - Short/Long-Term Memory (S/LTM)
 - Static vs dynamic maps
 - Identification of moving elements
 - Distinction between static – semi-static – dynamic changes
 - Ignore dynamic elements for static map
 - Use static map for localization
 - Multi-hypothesis framework

(not possible to show examples due to require payment of copyright fees...)

Long-term SLAM in Dynamic Environments

- Map representation
 - Short/Long-Term Memory (S/LTM)
 - Static vs dynamic maps
- Multi-hypothesis framework
 - Group landmarks positions based on signature
 - Hypothesis likelihood ~ matched measurements vs missed detections
 - Assume uniquely identifiable landmarks

(not possible to show examples due to require payment of copyright fees...)

Long-term SLAM in Dynamic Environments

- Map matching
 - Observation vs map
 - Clustering properties of observations
 - Ray tracing
 - NDT + occupancy grid map hybrid representation

Long-term SLAM in Dynamic Environments

- Prediction modeling
 - Markov-based processes
 - State holding time associated to Markov processes
 - Patterns periodicity
 - Fourier analysis
 - Time series as stationary stochastic processes
 - CNN-based prediction

Long-term SLAM in Dynamic Environments

- Dynamic objects detection
 - Assume certain semantic classes as dynamic
 - Use features associated to static objects for localization and mapping
 - Bounding box segmentation (YOLO)
 - Pixel-wise segmentation



Open Challenges

- Heterogeneous sensor configurations
- Multi-modal maps
- Tightly-coupling of dynamics estimation into localization and mapping
- Online estimation of the observations' dynamicity
- Human-Machine Interfaces (HMI)

Outline

- Introduction
- Literature Review
- Thesis Statement
 - Problem Statement
 - Research Questions (RQ) and Hypotheses (H)
- Preliminary Developments
- Research Plan

Thesis Statement: Problem Statement

- Localize a mobile robot within dynamic indoor environment
- Decide how and when to update the map
- No assumption on the environment dynamics

Thesis Statement: Problem Statement

- Localize a mobile robot within dynamic indoor environment
- Decide how and when to update the map
- No assumption on the environment dynamics
- Mathematical formulation:
 - $p(x_t, m_t, d_t | z_{1:t}, u_{1:t})$ (online SLAM)
 - $p(x_t, m_t, c_t, d_t | z_{1:t}, u_{1:t})$
 - $p(x_{1:t}, m_{1:t}, d_{1:t} | z_{1:t}, u_{1:t})$ (full SLAM)
 - $p(x_{1:t}, m_{1:t}, c_{1:t}, d_{1:t} | z_{1:t}, u_{1:t})$

Research Questions (RQ) and Hypotheses (H)

- **RQ1:** Extrinsic parameter calibration without external equipment
 - *H1: Convergence on relative motion of each sensor in the same coordinate frame should be maximum for the true value of the sensors' extrinsic parameters*
- **RQ2:** Improve mapping results and their long-term stability when performing SLAM in a dynamic indoor environment
 - *H2.1: Dynamics history in an area, the points state (position, velocity), their existence throughout time, and dynamics clustering properties relative to neighborhood should give a model for the environment dynamics*
 - *H2.2: Tightly-coupling of the environment dynamics modeling into mapping process using the dynamics as weights while also rejecting highly dynamic observations, giving more importance to static observations, should improve mapping results*

Research Questions (RQ) and Hypotheses (H)

- **RQ3:** Integrate environment dynamics into the localization process
 - *H3: Similar to H2.2, tightly-coupling of environment into the localization estimation process should achieve long-term stability of the localization results in a dynamic indoor environment*

Research Questions (RQ) and Hypotheses (H)

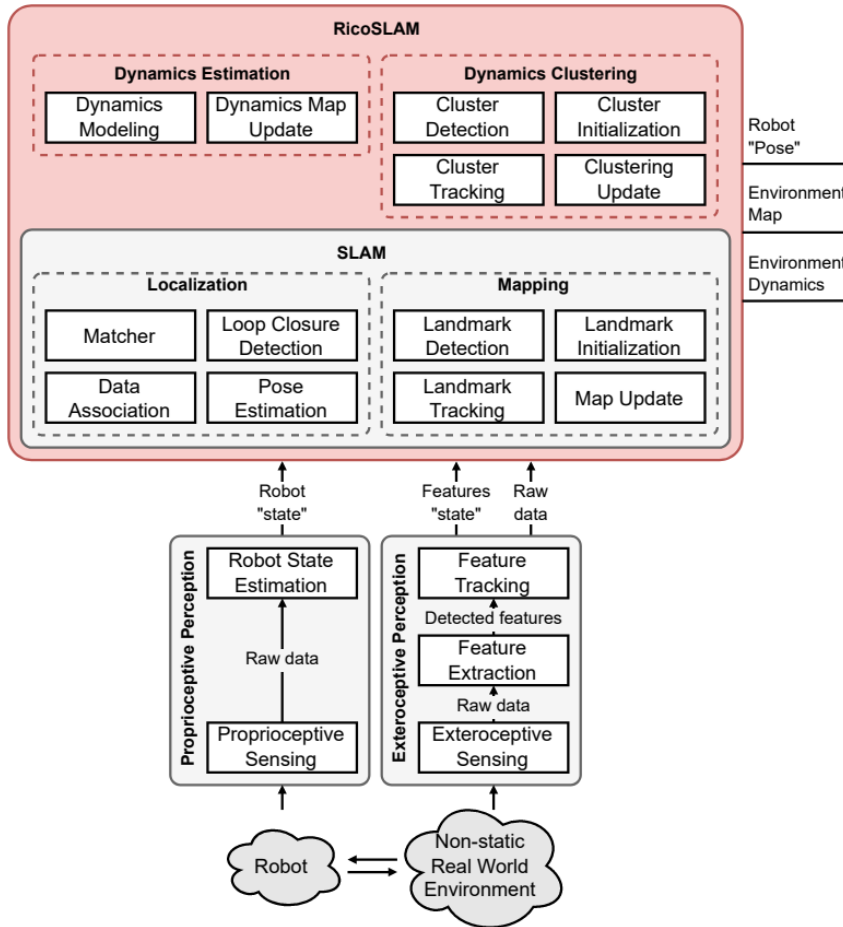


Fig. 6: Generic block diagram for this PhD thesis research proposal in the context of SLAM

Outline

- Introduction
- Literature Review
- Thesis Statement
- Preliminary Developments
- Research Plan

Preliminary Developments

- Preparation of the mobile platforms for data acquisition
- Preliminary study on the Hypothesis H1 for automatic robot calibration
- Introduction to SLAM

Outline

- Introduction
- Literature Review
- Thesis Statement
- Preliminary Developments
- Research Plan

Research Plan

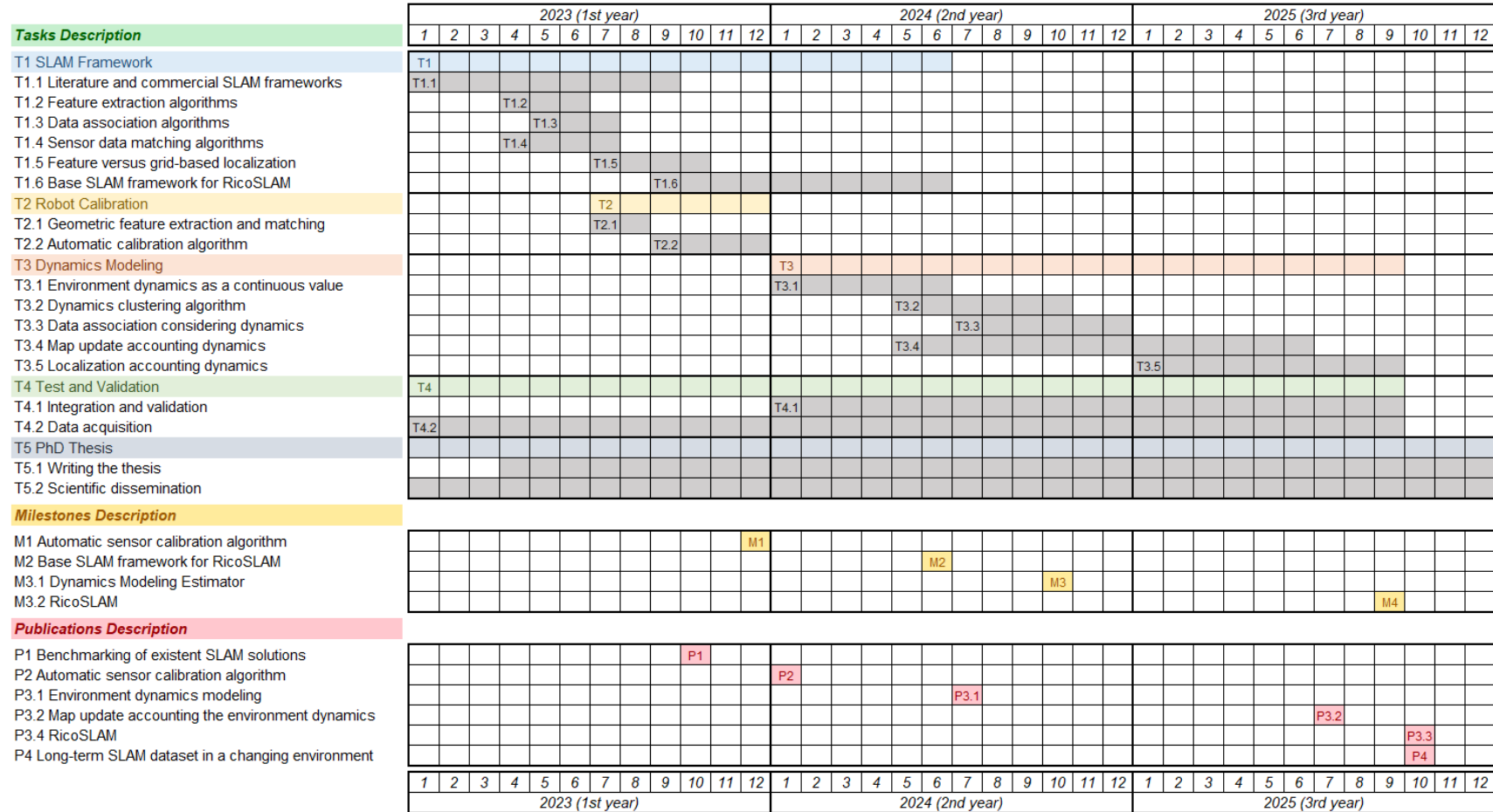


Fig. 7: Timeline of the PhD research plan

Research Plan

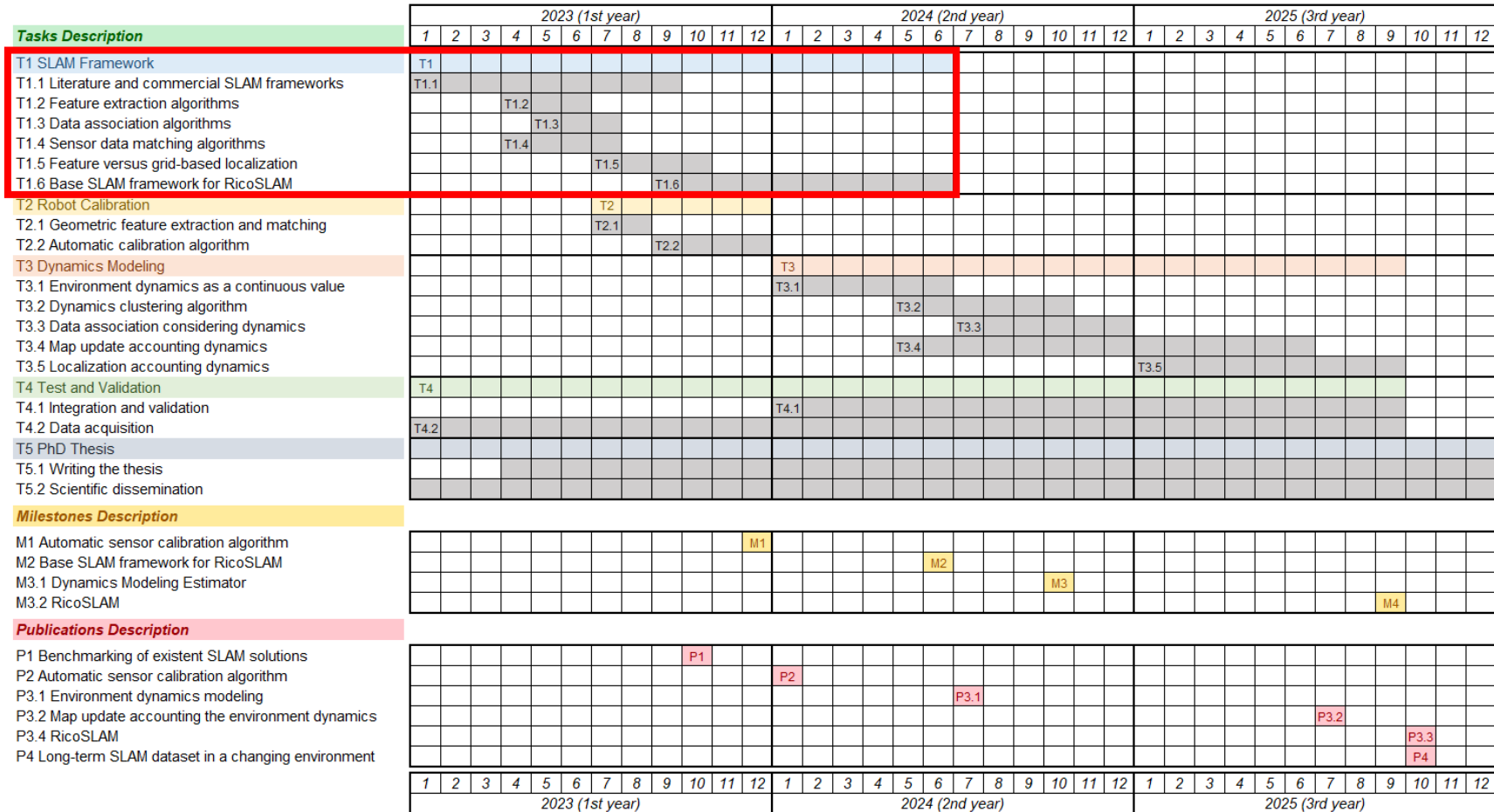


Fig. 7: Timeline of the PhD research plan

Research Plan

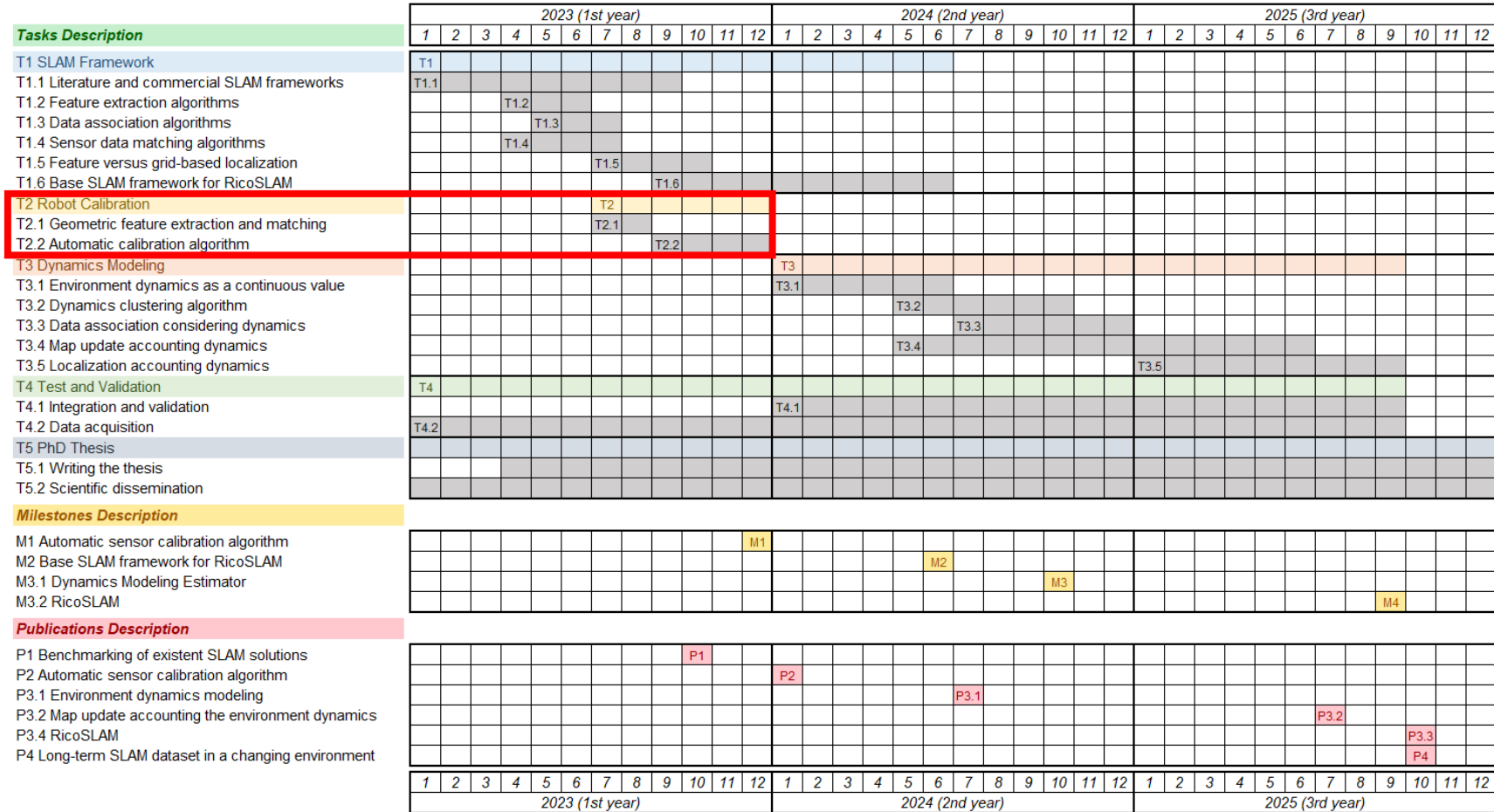


Fig. 7: Timeline of the PhD research plan

Research Plan

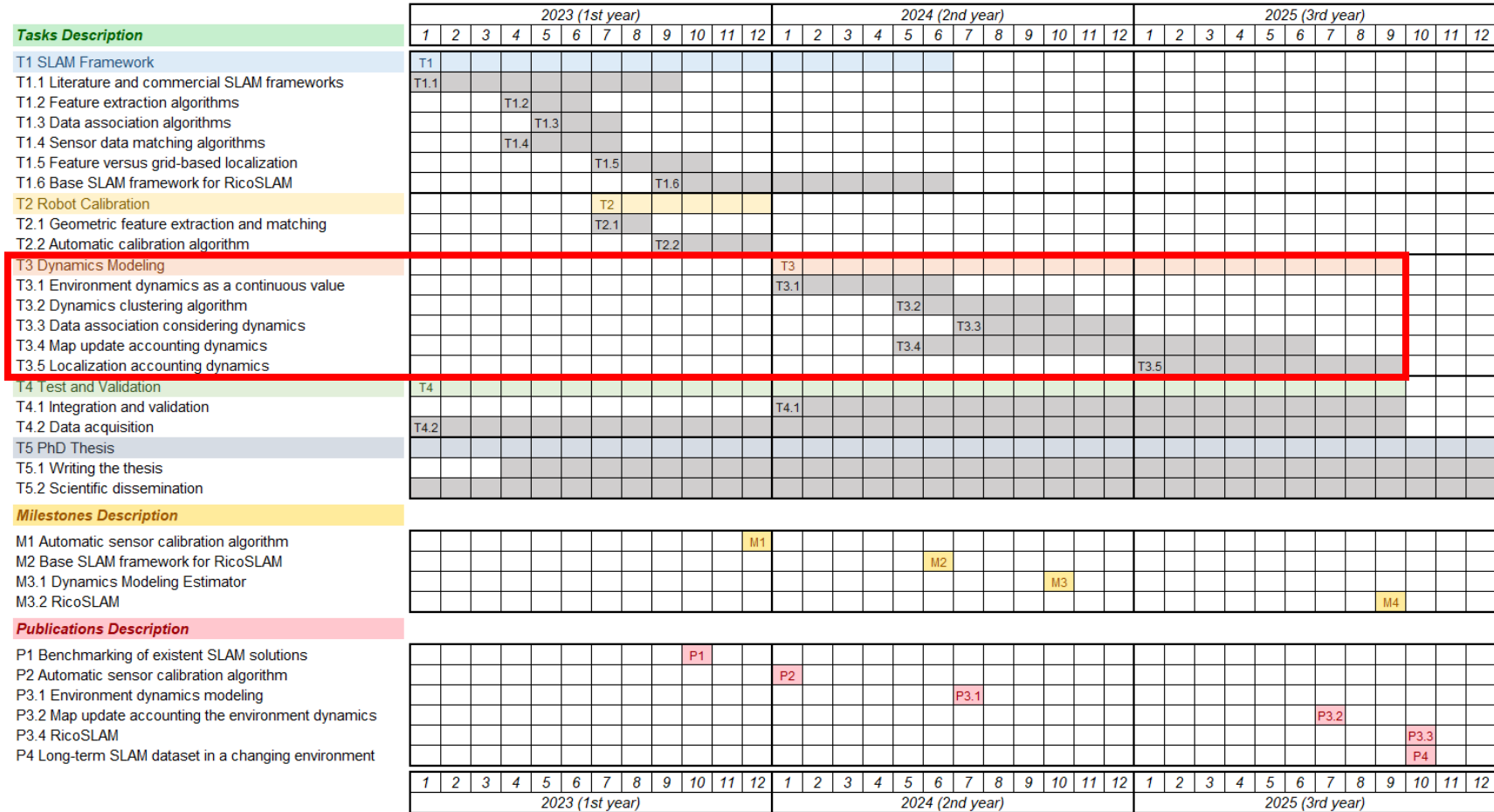


Fig. 7: Timeline of the PhD research plan

Research Plan

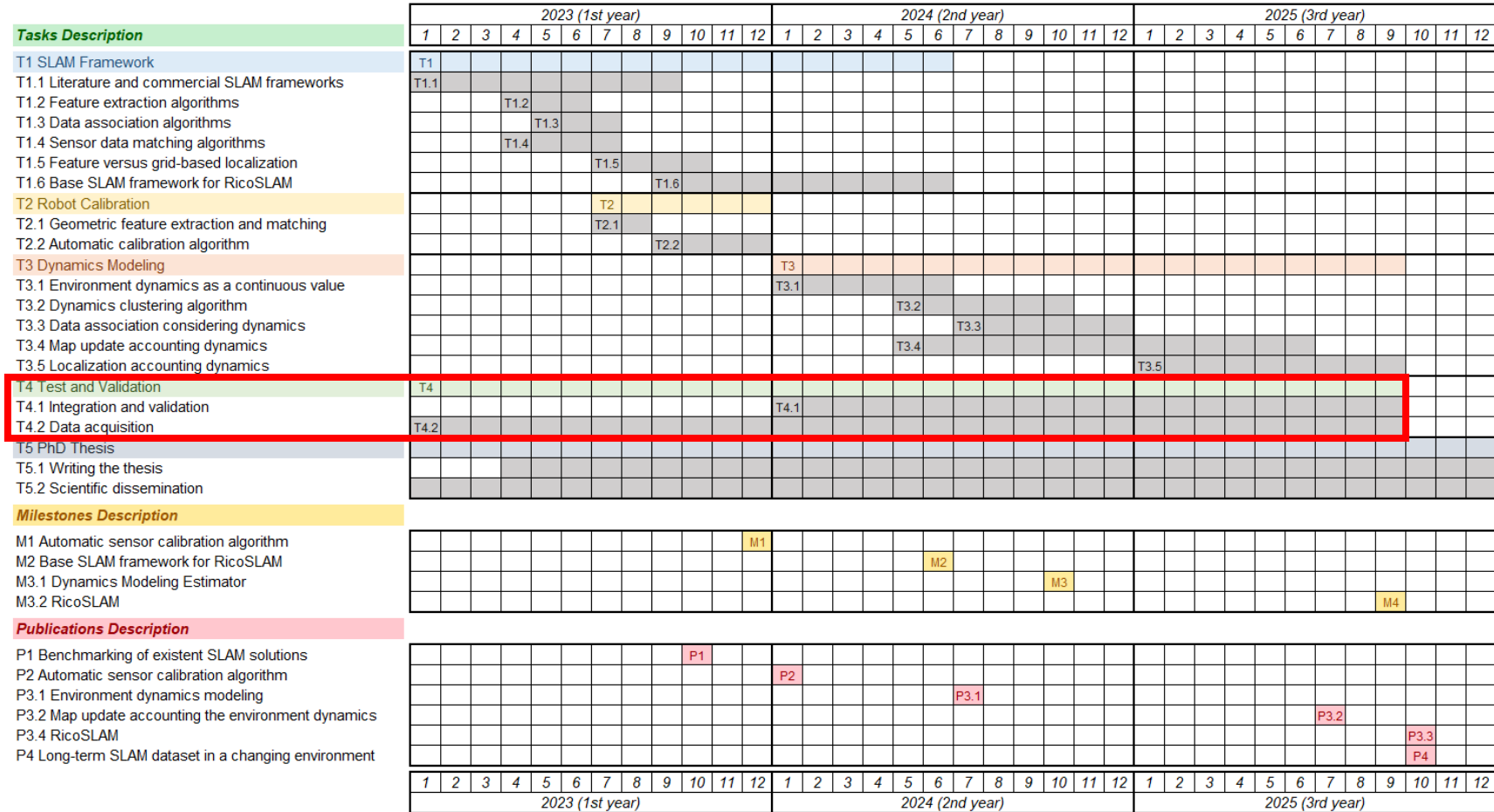


Fig. 7: Timeline of the PhD research plan

Research Plan

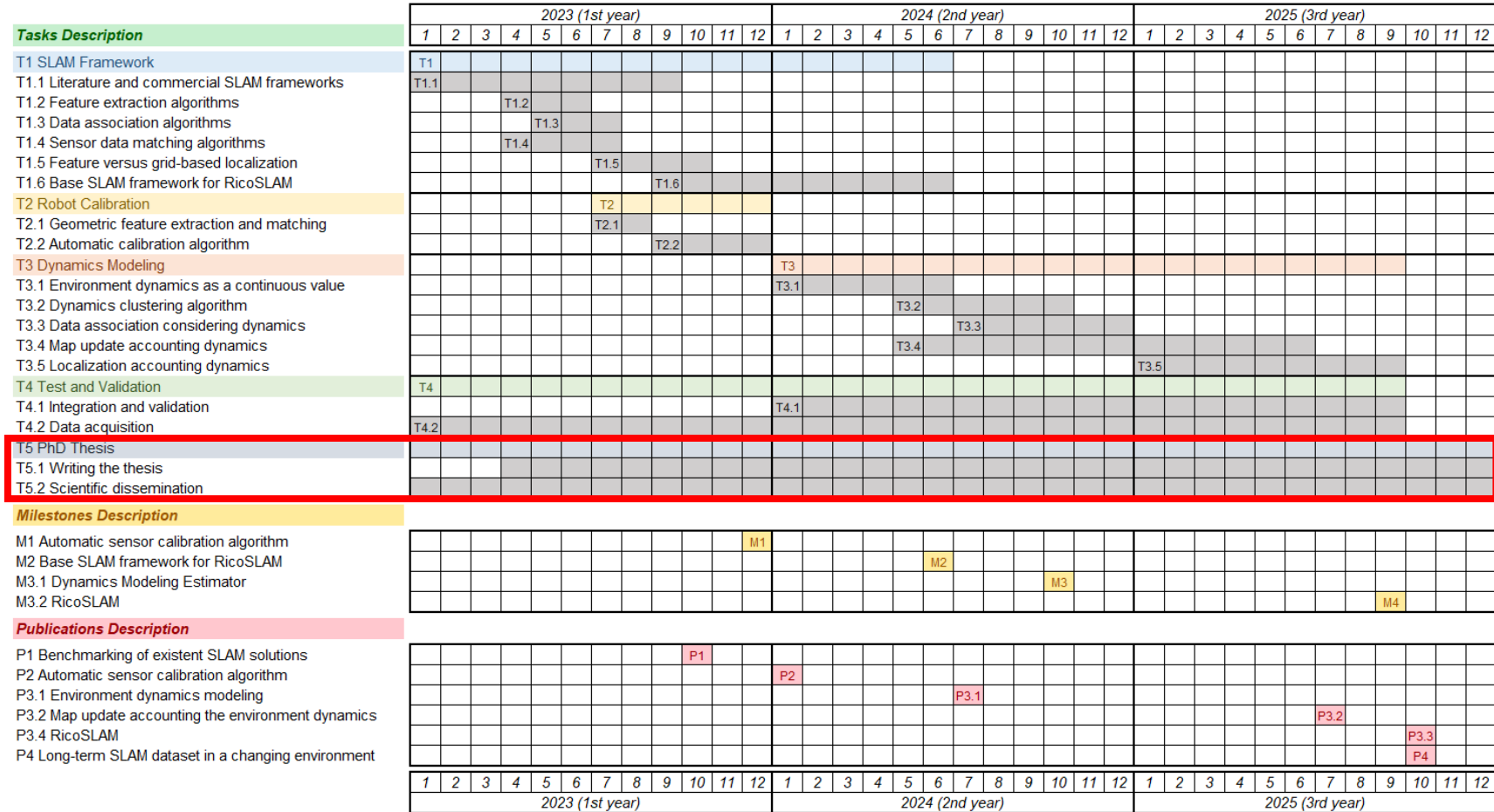


Fig. 7: Timeline of the PhD research plan

Research Plan

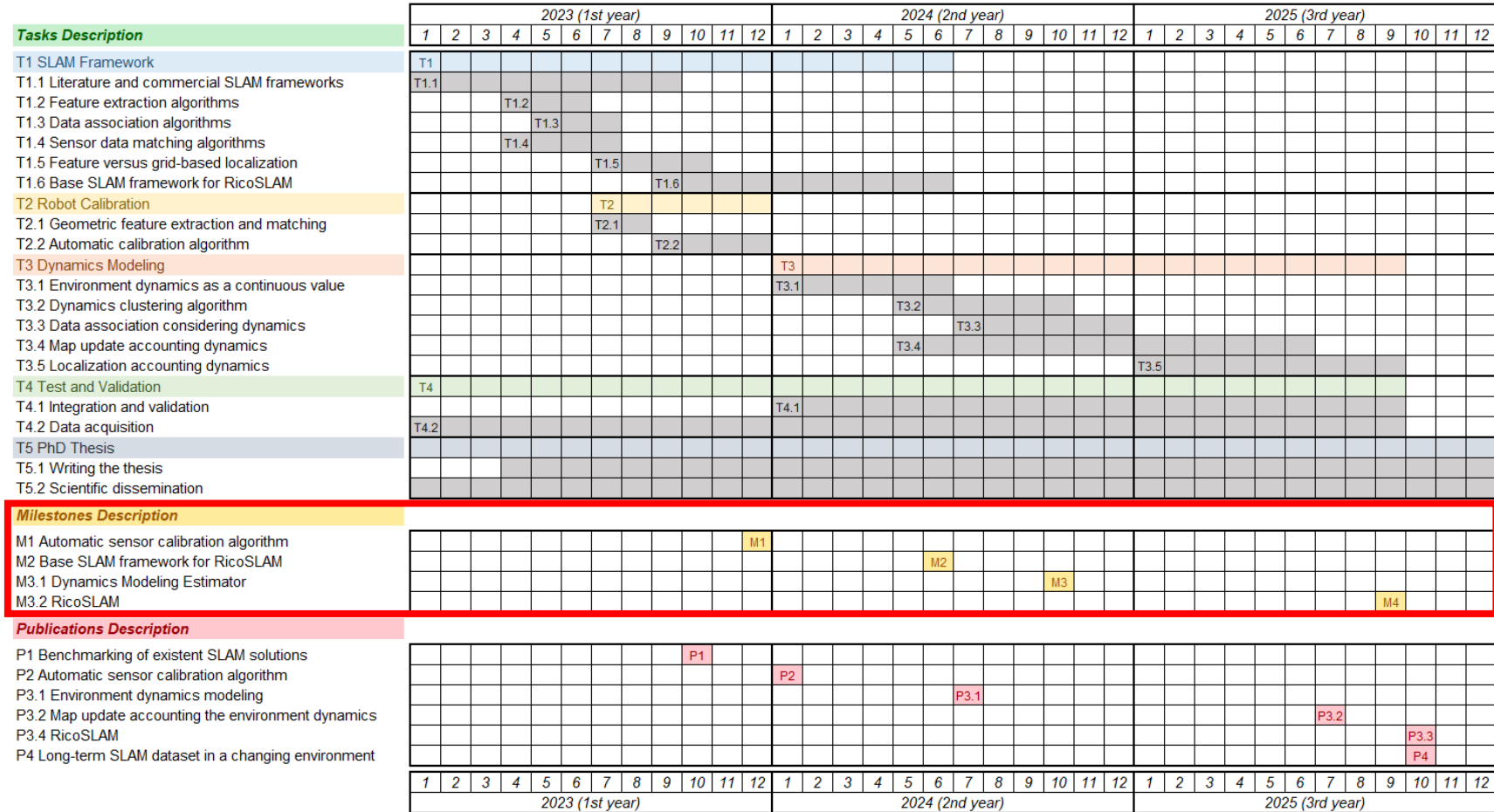


Fig. 7: Timeline of the PhD research plan

Research Plan

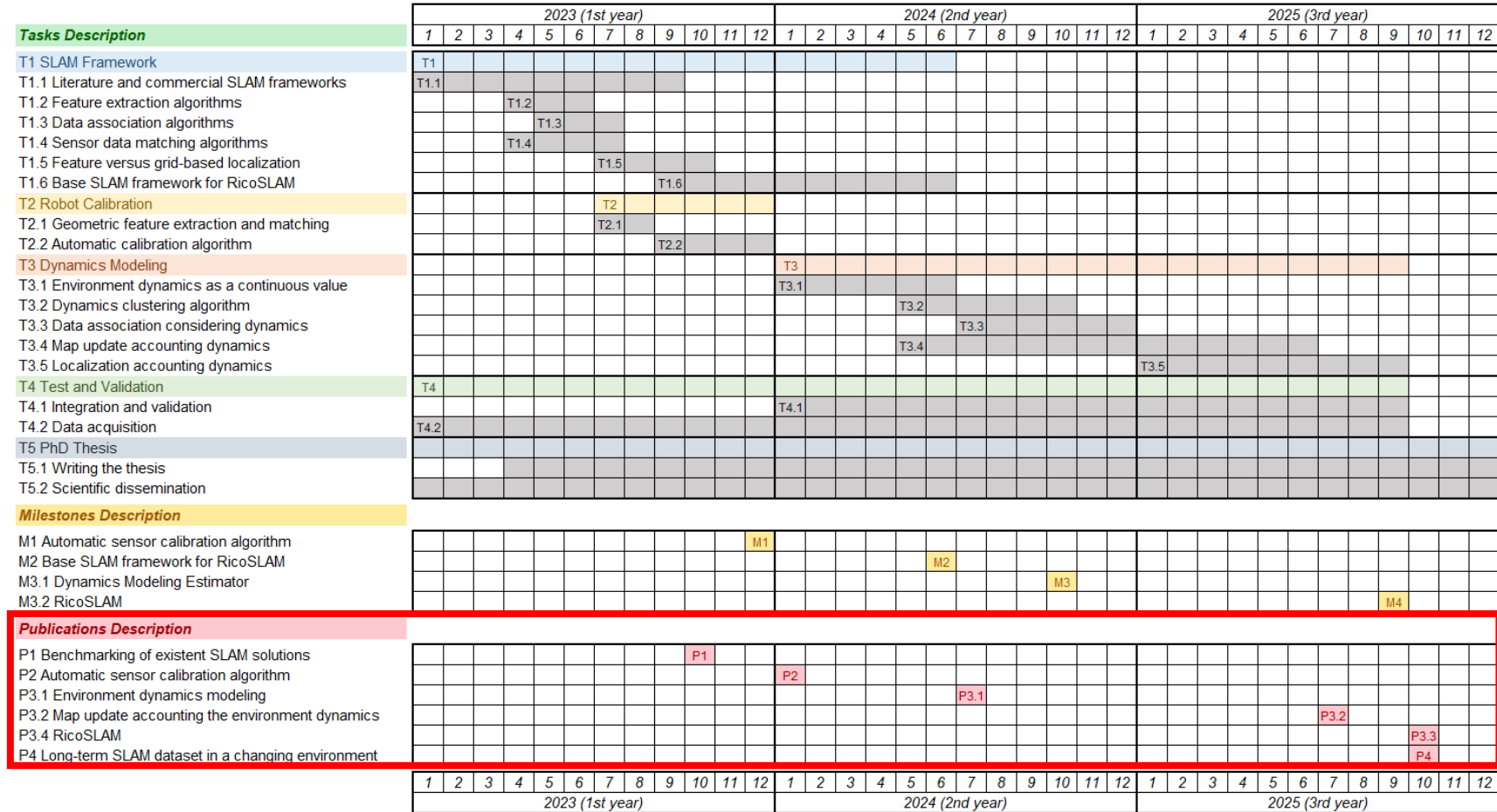


Fig. 7: Timeline of the PhD research plan

Research Plan

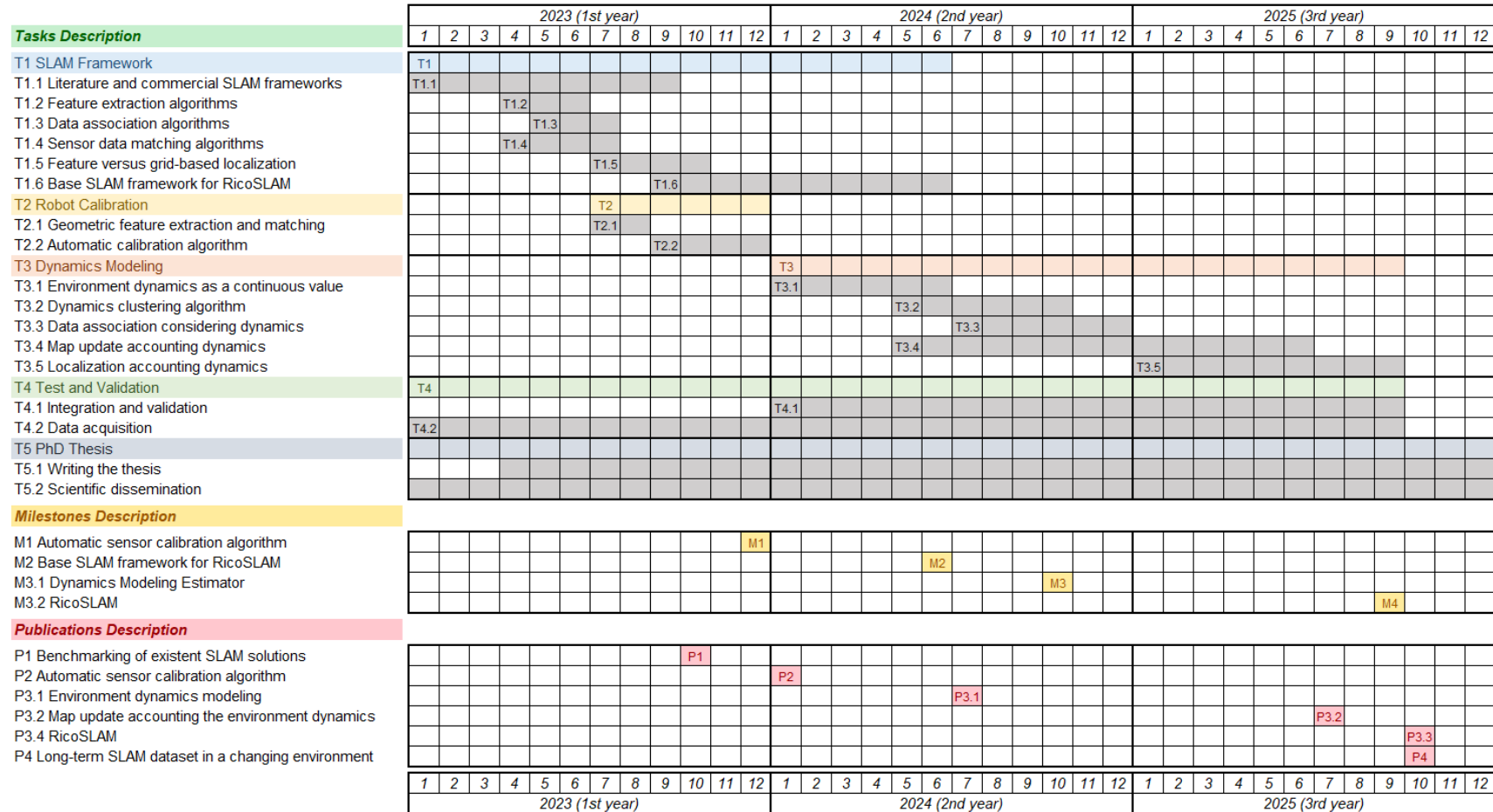


Fig. 7: Timeline of the PhD research plan

Any questions?

Ricardo B. Sousa

Thesis Research Plan – Doctoral Program in Electrical and Computers Engineering
Zoom Meeting, January 12, 2023

Prof. António Paulo Moreira, PhD (Supervisor)
Héber Miguel Sobreira, PhD (Second Supervisor)

RicoSLAM: Long-Term Localization and Mapping in Dynamic Environments

Ricardo B. Sousa

Thesis Research Plan – Doctoral Program in Electrical and Computers Engineering

Zoom Meeting, January 12, 2023

Prof. António Paulo Moreira, PhD (Supervisor)

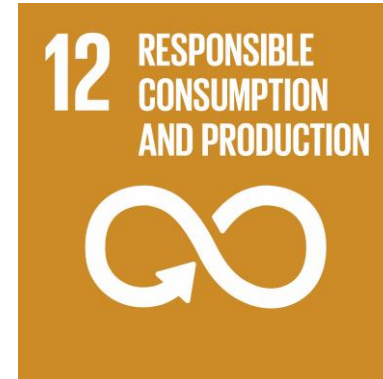
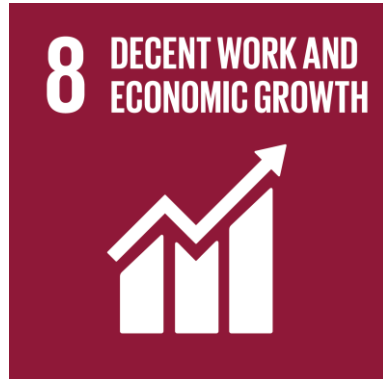
Héber Miguel Sobreira, PhD (Second Supervisor)

References

- [1] AgileIntel Research. Global autonomous mobile robot market – trends, COVID-19 impact and growth forecasts to 2029. Tech. rep. AgileIntel Research, May 2022. URL: <https://www.agileintelresearch.com/reportdetails/Global-Autonomous-Mobile-Robot-Market/43>.
- [2] Thomas Hahn, Morten Irgens, Bernd Liepert, Barry O’Sullivan, Arnold WM Smeulders, David Bisset, Edward Curry, Michela Milano, and Sonja Zillner. Strategic Research, Innovation and Deployment Agenda (SRIDA). Tech. rep. Big Data Value Association (BDVA), Confederation of Laboratories for Artificial Intelligence Research in Europe (CLAIRE), European Lab for Learning & Intelligent Systems (ELLIS), European Association for Artificial Intelligence (EurAI), and euRobotics, Sept. 2020. URL: <https://adr-association.eu/wp-content/uploads/2020/09/AI-Data-Robotics-Partnership-SRIDA-V3.0-1.pdf>.
- [3] United Nations. Transforming our world: the 2030 agenda for sustainable development. Last accessed on Oct. 17, 2022. Sept. 2015. URL: <https://sdgs.un.org/publications/transforming-our-world-2030-agenda-sustainable-development-17981>.
- [4] S. Thrun, W. Burgard, and D. Fox. Probabilistic robotics. The MIT Press, 2005. ISBN: 978-0-262-20162-9.
- [5] Ken Peffers, Tuure Tuunanen, Marcus A. Rothenberger, and Samir Chatterjee. “A design science research methodology for information systems research”. In: Journal of Management Information Systems 24.3 (2007), pp. 45–77. DOI: 10.2753/MIS0742-1222240302.

Introduction: Goals

- 2030 Agenda for Sustainable Development by the United Nations (UN) [3]



Introduction: Research Methodology

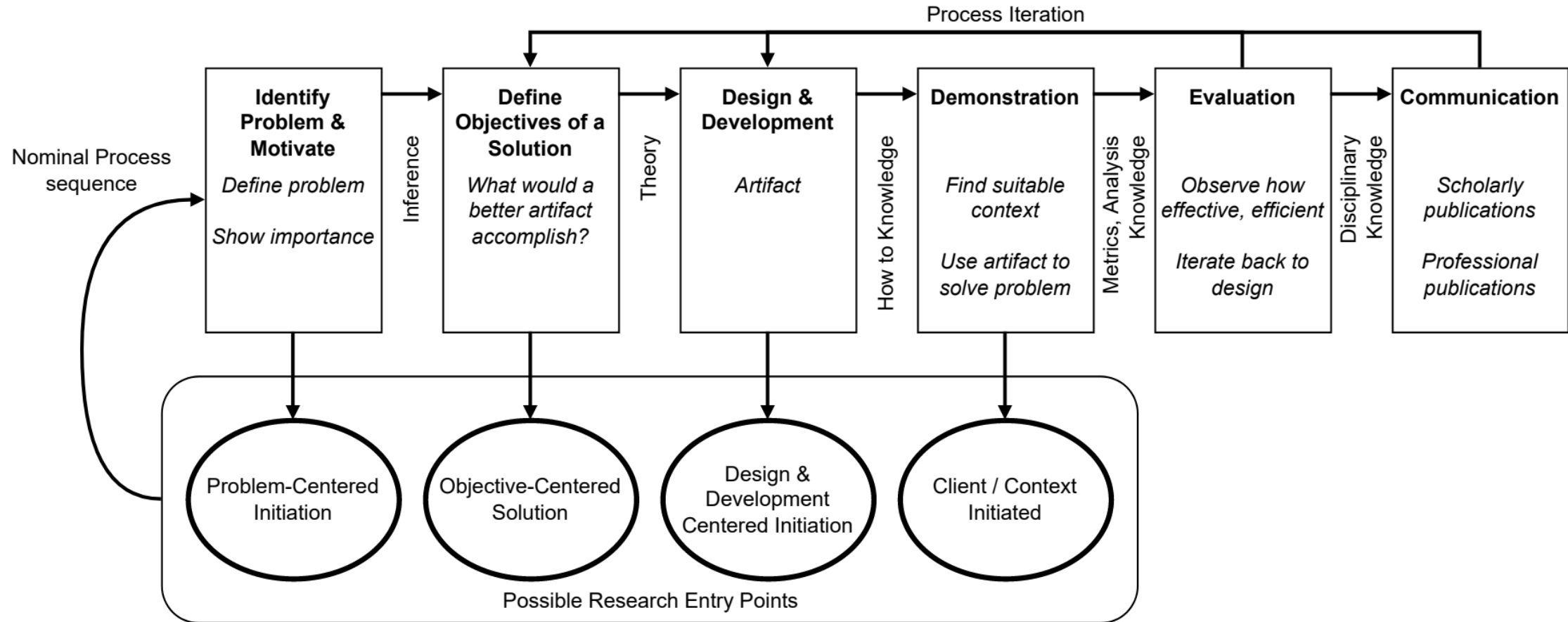


Fig.: Process model of the Design Science Research Methodology (DSRM) [5]

Outline (main topics)

- Introduction
- Literature Review
- Thesis Statement
- Preliminary Developments
- Research Plan

Outline (estimated times per topic)

- Introduction (~3min)
 - Context (1.5min)
 - Motivation (1min)
 - Goals (0.5min)
- Literature Review (~6min)
 - Methodology (1.5min)
 - SLAM (1min)
 - Long-term SLAM in Dynamic Environments (3.5min)
- Thesis Statement (~3min)
 - Statement (1min)
 - Q+H (2min)

Outline (estimated times per topic)

- Preliminary Developments (~1min)
- Research Plan (~2min)
 - Tasks
 - Milestones
 - Publications